

DESIGN VERIFICATION REPORT

Designer/ Manufacturer : Special and Safety Work (SSW), Riyadh, Saudi Arabia.
 DNV GL Project No : PP125007
 Equipment Description : Blast Resistant Window
 Codes and Standards : AISC (9th Edition)

This is to confirm that the structural design of the Blast Resistant Window (as per drawings/documents listed in Table-1:Design Examination Document) has been reviewed and found to be in compliance with the applicable codes and standards mentioned in this Design Verification Report.

Scope of Verification:

The verification of the design of window is limited to review of structural analysis submitted in compliance with AISC (9th Edition) as applicable with regards to local in-place strength against uniformly distributed static equivalent blast pressure of 2 psig.

No other codes / standards have been referred to and it is concluded that client would satisfy themselves that requirements of other relevant codes / standards / national regulations e.g. those mentioned as reference in design calculations submitted etc., are complied with suitably.

DNV GL review is limited to the structure of window itself.

Table-1: Design Examination Document:

S. No.	Drawing/Document No.	Revision	Drawing/Document Title	Status
1.	FSC-03	03	Blast Resistant Window-Elevation and Sections	Reviewed
2.	Blast Resistant Glazing Design 2007 Dated 25/Dec/2014	Version 1.6	Blast Resistant Glazing Report	FI*

FI* - For Information

Verification Limitations:**Design Parameters:**

Parameter	Description
Blast Pressure	2 psig
Overall Window frame Size	2000 mm width and 1200mm height
Rebound Pressure adopted	Nil

Other design data are considered as per drawing listed in Table-1.



Limitations and Notes:

- (1) The design appraisal is limited to the static in place analysis of the steel window frame and any aspects towards adequacy of anchoring arrangement, inspection and testing shall be ensured by the manufacturer.
- (2) The steel frame is found suitable to withstand the pressure load and the concrete part is excluded from our scope.
- (3) Blast Resistant Glazing Design report is used for information only.
- (4) Window structure steel sheet must have yield strength not less than 250 MPa as used in design calculation. All material used for window and the hardware components like window frame anchors, etc. including sealants along with fabrication and installation arrangements shall meet the blast rating specifications and structural quality as per the client requirements.
- (5) Present appraisal has been carried out with the blast load indicated in the submitted calculations. Other loads/ load factors like brittle fracture, etc are not considered in this review.
- (6) DVR revised to current revision in view of addition in for information document and change in Limitations and notes.

Validity Period:

No validity period is stated on the design verification report as it only confirms the conformity with specified regulations / standards at the time of issue. Change in regulations or change in design may have influence on the validity.

Signed for **Det Norske Veritas AS, Abu Dhabi Branch**



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Balaji Selvaraj
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OPMAE 646

Date: 18th March 2015
Place: Dubai, UAE